ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY 6031/3

PAPER 3

JUNE 2023 SESSION 2 hours 30 minutes

Additional materials:
- Answer paper
- Data Booklet
- Electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from Section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 12 printed pages.

Section A

Answer any two questions from this section.

1 (a) Manganese can be extracted from manganese (IV) oxide by reduction with carbon.

(i) Write an equation for the reduction process.

(ii) Calculate the enthalpy change for the extraction of manganese given that the enthalpy changes of formation of manganese (IV) oxide and carbon dioxide are $-520.3\ kJmol^{-1}$ and $-393.3\ kJmol^{-1}$ respectively.

(iii) Comment on the relative stability of manganese. [4]

(b) At 300 °C, gaseous phosphorous (V) chloride decomposes to give phosphorous (III) chloride gas, the process reaching an equilibrium.

(i) Write an equation for the decomposition process.

(ii) A sample of 0.12 moles of phosphorous (V) chloride in a closed vessel of 3 500 cm^3 was found to have 0.048 moles of phosphorous (III) chloride at equilibrium.

Calculate the

1 equilibrium concentrations of all components in the mixture,

2 K_c at this temperature. [6]

(c) The first six ionisation energies of an element **R** are shown in **Table 1.1**.

Table 1.1

ionisation number	1	2	3	4	5	6
ionisation energy / $kJmol^{-1}$	1000	1840	2880	4900	7820	16700

(i) Deduce, with reasons, the group of the periodic table to which element **R** belongs?

(ii) Give the electronic configuration of the outer most shell of **R**.

(iii) Draw diagrams to show the shapes of orbitals in the valency Shell of **R**.

[5]

[Total :15]

2 (a) When a 3.00 g sample of a mixture of $NaHCO_3$ and $NaCl$ was heated, the mass decreased by 0.31 g.

(i) Write a balanced chemical equation, for the reaction that occurred.

(ii) Calculate the mass of

1 $NaHCO_3$ in the sample,

2 $NaCl$ in the sample.

[5]

(b) At 30 °C, a solution of NaOH had a pH of 12.50.

(i) Calculate the concentration of NaOH given that K_w at 30 °C is 1.4×10^{-14} mol^2dm^{-6}.

(ii) Identify any **one** factor that affects K_w.

[5]

(c) Nickel is extracted from its ores by carbon reduction. The impure nickel produced is purified by electrolysis.

(i) Name any **one** metal element found in impure nickel.

(ii) Describe the electrolytic purification of the impure nickel.

[5]

[Total:15]

3 (a) The reaction between potassium manganate (VII), $KMnO_4$ and oxalic acid, $H_2C_2O_4$, is autocatalysed.

The oxidation half equation for the reaction is:

$$H_2C_2O_4 \rightleftharpoons 2CO_2 + 2e^- + 2H^+$$

(i) Name, with reasons, the type of reaction.

(ii) Define the term *autocatalysis*.

(iii) Sketch a graph to show how the rate of the reaction in (i) changes with time.

(iv) A 0.275 g sample of pure $H_2C_2O_4$ crystals were dissolved in dilute H_2SO_4, to which 0.2 g of pure $KMnO_4$ was added.

Calculate the volume of CO_2 produced at r.t.p.

[Use of relevant data from the data booklet is recommended] [7]

(b) The kinetics of a reaction involving three reactants, **A**, **B** and **C**, gave the results shown in **Table 3.1**.

Table 3.1

Experiment No.	**Initial [A] / moldm^{-3}**	**Initial [B] / moldm^{-3}**	**Initial [C] / moldm^{-3}**	**Initial rate / moldm^{-3}**
1	0.2	0.02	0.002	1.8×10^{-3}
2	0.2	0.04	0.002	3.6×10^{-3}
3	0.3	0.04	0.002	5.4×10^{-3}
4	0.3	0.02	0.004	2.7×10^{-3}

(i) Deduce the order of reaction with respect to the concentration of

1 **A**,

2 **B**,

3 **C**.

(ii) Calculate the rate constant.

(iii) Write the rate equation.

[8]

[Total:15]

Section B

*Answer **one** question from this section.*

(a) (i) Define the term *inert pair effect*.

(ii) When a brown Group (IV) compound, **G**, is strongly heated, it turns yellow – orange. Addition of concentrated HCl to **G** produces a white solid and a gas, **H**, that bleaches litmus paper.

1 Identify **G** and **H**.

2 Write chemical equations for the chemical changes described. [5]

(b) (i) Differentiate between the reaction of Cl_2 and H_2 from that of I_2 and H_2.

(iii) Compare the oxidising abilities of Br_2 and I_2 in terms of E^{θ} values.

[Use of Fe^{3+} / Fe^{2+} half equation is recommended]

(iii) Iodide ions are found in small quantities in iodised salts. Some foods salted with iodised salt turn yellowish when left to stand in air for a while.

Suggest an explanation for the observation in terms of E^{θ} values. [10]

[Total:15]

5 (a) (i) Describe, with the aid of chemical equations, the acid-base nature of the oxides of magnesium, aluminium and silicon.

(ii) Describe and explain the variation in the second ionisation energies of Period 3 elements, Na to Ar.

[8]

(b) (i) Write balanced chemical equations for the

1 oxidation of ammonia by atmospheric oxygen,

2 formation of nitric acid from nitrogen dioxide.

(ii) Explain how the addition of lime affects ammonium sulphate fertilisers in the soil.

(iii) Describe and explain how the accumulation of nitrate ions in drinking water sources affects human health.

[5]

(c) Ammonium nitrate, (NH_4NO_3), decomposes to give nitrogen gas and other gaseous products.

(i) Write an equation for the decomposition.

(ii) Suggest a use of ammonium nitrate based on this reaction.

[2]

[Total:15]

Section C

*Answer any **two** questions from this section.*

(a) A 1.00 g sample of hydrocarbon, **Q**, was vapourised at 77 °C and 100 kPa. The gas produced occupied a volume of 220 cm^3.

(i) Calculate the relative molecular mass of **Q**.

(ii) Deduce the molecular formula of **Q**, given that its empirical formula is C_5H_6. [5]

(b) When **Q** was refluxed with acidified concentrated potassium manganate (VII), a mixture of two carboxylic acids was obtained. One of the acids has the structural formula shown in **Fig. 6.1**.

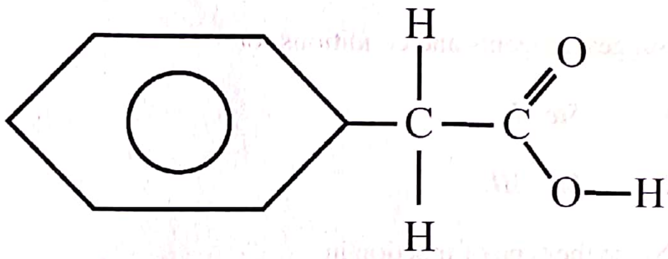


Fig. 6.1

(i) Describe the observation made when the concentrated acidified $KMnO_4$ is added to **Q**.

(ii) State the type of reaction that occurs between **Q** and acidified $KMnO_4$.

(iii) Suggest the formula of the other acid.

(iv) Deduce the structural formula of **Q**.

(v) Draw the structural formula of a hydrocarbon which is isomeric with **Q**. [7]

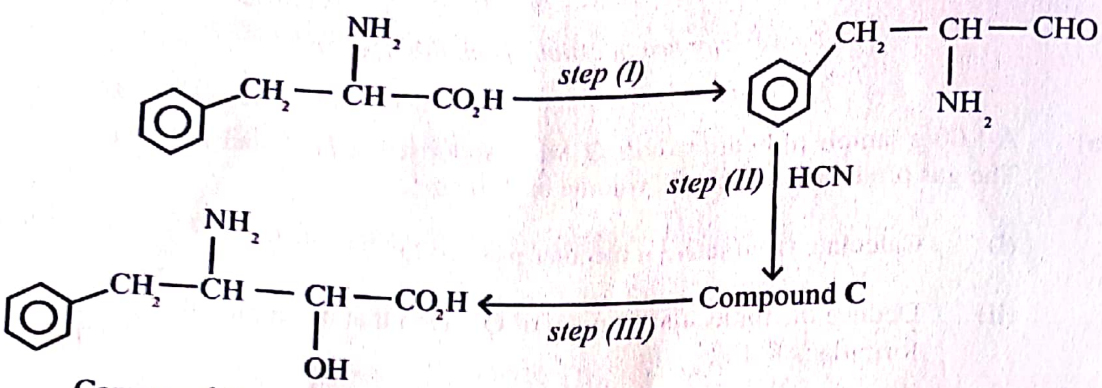
(c) (i) Define the term *polymerisation*.

(ii) Name the type of polymerisation that **Q** undergoes.

(iii) Draw a segment of a polymer made from **Q**. [3]

[Total:15]

7 (a) The reaction scheme shown in **Fig. 7.1** shows how an organic compound, **B**, can be synthesized.



Compound **B**

Fig. 7.1

(i) State the number of chiral centres in **B**.

(ii) Suggest reagents and conditions for

1 *Step I*,

2 *Step III*.

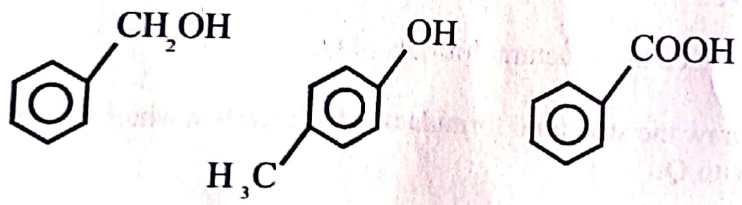
(iii) Name the type of reaction in

1 *Step I*,

2 *Step III*.

(iv) Give the structure of **C**. [6]

(b) **Fig. 7.2**. shows the structural formulae of three organic compounds **P**, **Q** and **R**.



P **Q** **R**

Fig. 7.2

(i) Explain the relative acidities of **P**, **Q** and **R**.

(ii) Describe a chemical test that can be used to distinguish **R** from **P** and **Q**. [6]

(c) The structure of an organic compound **M**, is shown in **Figure 7.3**.

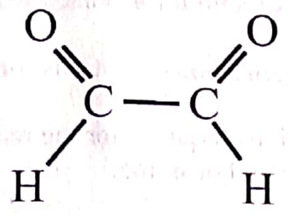


Fig. 7.3

(i) Give the structure of the organic product formed when **M** reacts with

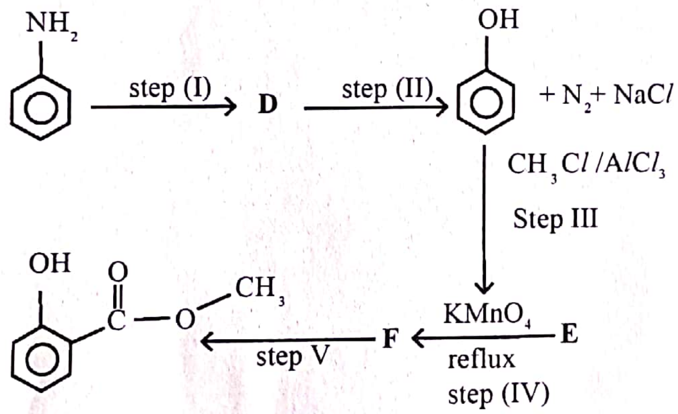
1 $LiAlH_4$,

2 2,4 - dinitrophenylhydrazine.

(ii) Name the type of reaction undergone by **M** when it reacts with 2,4 - dinitrophenylhydrazine. [3]

[Total:15]

8 **Fig. 8.1** shows how an organic compound, **G** can be synthesised.



Compound **G**

Fig. 8.1

(a) Name compound **G**. [1]

(b) Compounds **Y** and **Z** are isomers of **G** and they react with PCl_5. **Y** reacts with PCl_5 in the ratio 1:1 and **Z** in the ratio 1:2.

(i) Deduce the structural formulae of **Y** and **Z**.

(ii) Name the type of structural isomerism shown.

(iii) State the observations made when **G** reacts with PCl_5? [4]

(c) (i) Deduce the structural formulae of **D**, **E** and **F**.

(ii) Suggest reagents and conditions for *steps I*, *II* and *V*. [6]

(d) (i) State conditions required for the reaction between methyl cyclohexane and bromine.

(ii) Give a mechanism for the reaction. [4]

[Total:15]

Section D

Answer **one** question from this section.

9 **(a)** Define the term

(i) *partition*,

(ii) *partition coefficient*. [3]

(b) State two conditions under which a partition coefficient varies. [2]

(c) State the oxidation state of the transition element in

(i) $VO_{2\,(aq)}^{+}$,

(ii) $MnO_{4\,(aq)}^{-}$. [2]

(d) **(i)** Predict the reactions, if any, that occur when acidified solutions containing $MnO_{4\,(aq)}^{-}$ and $I_{(aq)}^{-}$ are mixed.

(ii) Suggest the effect of adding Fe^{3+} / Fe^{2+} to the reaction mixture. [8]

[Total:15]

10 (a) One of the requirements by the Environmental Management Agency (EMA) is that industries should have holding sites for their effluent before the effluent is released to the environment.

(i) Give **two** advantages of having holding sites.

(ii) Name any **two** general pollutants in the mining industrial effluent.

(iii) State the effect of each of the pollutants named in **(ii)**.

(iv) Describe ion exchange treatment of industrial effluent. [9]

(b) Carbon nanotubes, (CNTs), can be used in medicine for drug delivery and thermal therapy. Surface modifications are done to help CNTs to reach target sites.

(i) State **two** characteristic properties of CNTs that make them suitable for these uses.

(ii) Suggest why surface modifications are possible.

(iii) CNTs absorb near Infrared (NIR) radiation and emit it as heat.

Suggest an explanation of how this property can be exploited in thermal therapy. [6]

[Total:15]